

Ep = 18

when problem is large then divide it in multiple sub problems and solve them and combine them to get solution of main problem

To solve the recursion we need find what is smallest problem

→ it is type of recursive problem means sub problem should also a type of main problem then only it is a Divide and conquer strategy

→ General way.

```

DAC(P)
{
    if (small(P))
    {
        solve(P);
    }
    else
    {
        Divide P into P1, P2, P3 ... Pk
        Apply DAC(P1), DAC(P2) ...
        Combine (DAC(P1), DAC(P2) ...)
    }
}

```

→ where it is used

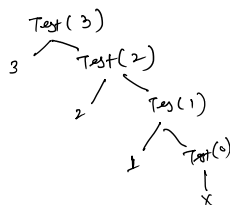
- ① Binary Search
- ② Finding Max and min
- ③ Merge Sort
- ④ Quick Sort
- ⑤ Strassen's matrix multiplication

Recurrence Relation

```

void Test(n)
{
    if(n > 0)
    {
        printf()
        Test(n-1)
    }
}

```



observation :- if $n=3$ we are calling it 4 times and 3 times in printing

if we pass n then it will call $n+1$ times and printf will be called n times

Time function = $f(n) = n+1$ call
 $O(n)$

How to solve recurrence relation

```

void Test(n) — T(n)
{
    if(n > 0)
    {
        printf() — ① unit of time
        Test(n-1) — T(n-1)
    }
}

```

$T(n) = T(n-1) + 1$

for Recurrence relation function name is T so we use $T(n)$

Recurrence Relation

$$T(n) = \begin{cases} T(n-1) + 1 & n > 0 \\ 1 & n = 0 \end{cases}$$

Now lets solve

$$T(n) = T(n-1) + 1$$

$$\therefore T(n-1) = T(n-2) + 1$$

$$T(n-2) = T(n-3) + 1$$

substitute $T(n-1)$

$$T(n) = [T(n-2) + 1] + 1$$

...

el again substitute

$$T(n) = T(n-3) + 3$$

k times

assume $n - K = 0$

$$\therefore n = k$$

$$T(n) = T(0) + n$$

$$T(n) = 1 + n$$

$O(n)$

$$O(n)$$

void Test(n) — $T(n)$

2 if($n > 0$)

```
for (i=0; i<n; i++) - n+1
```

```

{
    printf( n )
}

```

$$T_{\text{Test}}(n-1) \quad \text{---} \quad T(n-1)$$

3

$$T(n) = T(n-1) + (n) + (n+1) + 1$$

$$= T(n-1) + 2n + 2$$

To solve easily as $(2n+2)$ is linear
we are taking as linear value as n

$$T(n) = T(n-1) + n$$

$$T(n) = \begin{cases} 1 & n=0 \\ T(n-1) + n & n>0 \end{cases}$$

$$\begin{aligned} T(n) &= T(n-1) + n \\ T(n) &= (T(n-2) + (n-1)) + n \end{aligned}$$

$$T(n) = T(n-2) + 2n - 2$$

$$\tau(n-k) + n-k$$

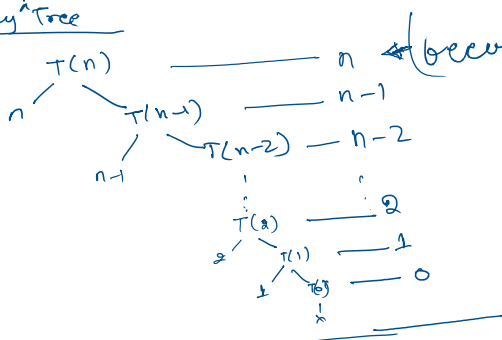
$$\text{Ex. } n - k = 0$$
$$n = k$$

$$\tau(n) = \tau(0) + k^2 - k$$

$$T(n) = 1 + k^2$$

done by me

Lets solve by ^{Recursion} Tree



$$= \frac{0+1+2+\dots+(n-2)(n-1)(n)}{2}$$

$$= \frac{n^2 + n}{2}$$

$$= \Theta(n^2) \text{ or } O(n^2)$$

lets solve by sub

$$T(n) = T(n-1) + n \quad \text{--- (2)}$$

$$T(n) = [T(n-2) + n-1] + n$$

$$= T(n-2) + (n-1) + n$$

$$= \left[\overset{\text{substitute}}{r(n-3)} + n - 2 \right] + (n-1) + n$$

$$= T(n-3) + (n-2) + (n-1) + n \quad \text{--- (3)}$$

lets do for K times

$$= 1(n-k) + (n-k-1) + (n-k-2) + \dots + (n-1) + n \quad \text{--- (4)}$$

Assume $n-k=0$
 $n=k$

$$\tau(n) = \tau(n-n) + (n-(n-1)) + \dots + (n-1) + n$$

$$= T(0) + 1 + 2 + \dots + (n-1) + \dots + (n)$$

$$= 1 + \left(\frac{n \times (n+1)}{2} \right)$$

$\Theta(n^2)$

$$T(n) = T(n-1) + n$$

$$T(n-1) = T(n-2) + n - 1$$

$$T(n-2) = T(n-3) + n-2$$

Ex - 2.1

Easy and Quick way

```

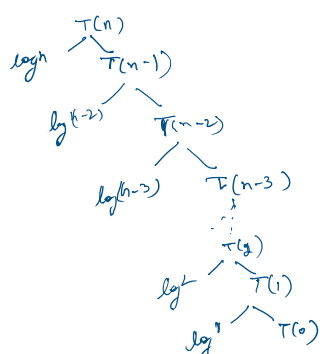
T(n) → void Test(n)
        ↓
        if (n > 0)
        ↓
        foo(i=0; i < n; i=i+2)
        ↓
        print()
        ↓
        Test(n-1)
        ↓
        }

```

$$T(n) = T(n-1) + \log n$$

relation

$$T(n) = \begin{cases} 1 & n=0 \\ T(n-1) + \log n & n>0 \end{cases}$$



$$\begin{aligned}
 &= \log n + \log(n-1) + \dots + \log 2 + \log 1 \\
 &= \log(n \times (n-1) \times (n-2) \times \dots \times 1) \\
 &= \log n!
 \end{aligned}$$

There is tight bound for this
upper bound of $n!$ is n^n

But $\log n!$ upper bound is $n \log n$
 $O(n \log n)$

Let's solve by substitution

$$T(n) = T(n-1) + \log n \quad \text{--- (1)}$$

$$T(n) = [T(n-2) + \log(n-1)] + \log n$$

$$= T(n-2) + \log(n-2) + \log n \quad \text{--- (2)}$$

$$= [T(n-3) + \log(n-3)] + \log n$$

$$= T(n-3) + \log(n-3) + \log(n-2) + \log n \quad \text{--- (3)}$$

$$= \vdots k$$

$$T(n) = T(n-k) + \log 1 + \log 2 + \dots + \log(n-1) + \log n$$

$$= T(0) + \log 1 + \dots + \log n$$

$$T(n) = T(0) + \log n!$$

$$= 1 + \log n!$$

$$O(n \log n)$$

Let's directly answer

$$\begin{aligned}
 T(n) &= T(n-1) + 1 && O(n) \\
 T(n) &= T(n-1) + n && O(n^2) \\
 T(n) &= T(n-1) + \log n && O(n \log n)
 \end{aligned}$$